Design and Development of Avatars for Digital Twin Music Concerts Using Metahuman Unreal Engine 5

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***Abstract*—This research aims to develop a digital twin musician in the form of a photorealistic avatar using Metahuman Unreal Engine 5 technology, which is integrated with motion capture, face recognition, and clothing simulation to create an immersive virtual concert experience. The methods used include the creation of 3D face models with HyperHuman AI and Polycam, clothing design using Marvelous Designer, and implementation of body movements and facial expressions through Move AI, DeepMotion, and Live Link Face. The results show that the avatars produced have high visual similarity, natural movements, and realistic facial expressions when performing on a virtual stage. Performance testing achieved a minimum target of 30 FPS at 1080p resolution, proving the feasibility of this technology for the digital music industry. In conclusion, the development of Metahuman-based digital twins opens up new opportunities in virtual entertainment, including NFT concerts and interactive digital merchandise.**

***Keywords*—Digital Twin, Metahuman, Motion Capture, Virtual Concert, Unreal Engine 5, Performance Optimization**

I. INTRODUCTION

The digital transformation of the music industry has evolved significantly over recent decades, from vinyl records to cassettes, CDs, and now streaming platforms with increasingly immersive digital experiences. As Ruddin et al. (2022) concluded, digitalization offers numerous opportunities to expand markets and increase profitability by reducing physical production costs and transitioning to digital platforms [1]. The rapid technological advancement enables music producers to market their best works virtually without requiring the physical presence of actual musicians. A notable example of this evolution is MAVE, a virtual K-pop group created using Unreal Engine and MetaHuman technology, demonstrating how advanced technology can produce highly realistic and engaging virtual musicians [2]. The presence of groups like MAVE not only creates new opportunities in music production but also in merchandise management. The music industry can now explore new business avenues in creating and marketing unique and interactive digital merchandise, extending beyond traditional physical merchandise limitations. The integration of blockchain technology, particularly through Non-Fungible Tokens (NFTs), presents an opportunity to create unique digital merchandise with verifiable authenticity and potential collectible value. Saputera et al. (2024) suggest that in the metaverse world, musicians and music producers have opportunities to form virtual NFT fictional music groups or bands, distribute music NFTs, sell NFT tickets, and sell band merchandise within the metaverse. NFTs can function as ”digital tickets” that provide access not only to physical concerts but also to exclusive virtual experiences in the metaverse. Unreal Engine 5 serves as the primary development platform providing a robust foundation for creating detailed and responsive virtual environments. One revolutionary feature integrated into Unreal Engine 5 is MetaHuman Creator, enabling the creation of hyperrealistic digital avatars with extraordinary detail. This technology accelerates the digital character creation process from months to mere hours [3]. This research aims to develop digital musician representations based on Metahuman with associated attributes and body movements to fulfill the requirements of musician performance on stage with a minimum target of 30 fps. The research is limited to the design and creation of Metahuman avatars with attached attributes and movements combined with Facial Motion Capture for concert simulation.

II. LITERATURE REVIEW

A. Previous Research

In previous research conducted by Ou, H., et al., ”Development of a low-cost and user-friendly system to create personalized human digital twin,” they focused on developing a system enabling individuals without art and design backgrounds to create their digital twins. Their approach addressed three main aspects: simplifying the digital character creation process through user-friendly interfaces, developing an integrated workflow for data-based digital character creation without specialized expertise, and creating a wearable sensor system capable of simultaneously capturing facial expressions, gaze direction, and body movements [5]. For face generation, their research used a 3D dense face alignment model mapping multiple facial depth points based on MobileNet with 3DMM

(3D Morphable Model) parameters. This model was selected for its ability to generate individual 3D face models from 2D face images with high speed and good fidelity. The system was implemented using PyTorch framework and accessed through a web-based interface developed with Streamlit. Unlike their approach, our research employs a combination of HyperHuman AI and Polycam, offering a more modern approach to generating accurate 3D facial meshes. For motion capture, their study used a wearable system comprising fifteen custom IMU sensors. Each IMU used BNO055 sensors to detect 3D acceleration, 3D angular velocity, and 3D magnetic field. This data was processed using an ESP32 Micro Control Unit for logging, processing, and wireless communication with PC-based applications. Their system could perform real-time motion reconstruction at a lower cost compared to commercial systems. Meanwhile, our research uses Move One, which is computer vision-based, eliminating the need for specialized hardware while still aiming to achieve accurate motion capture at an affordable cost. For facial tracking, both studies used the Live Link Face App for real-time facial expression capture. Ou et al. used an iPhone 13 mounted on a specialized headset for face and eye tracking. The use of Live Link Face in both studies indicates that this tool has become an industry standard for facial tracking in digital twin development. While both studies aim to create realistic digital twins, our research focuses specifically on implementation in the context of music concerts, whereas Ou et al.'s research was intended for telemedicine applications.

B. *Digital Twin in Music Industry*

Digital Twin represents a concept that has been known since the 1970s and has evolved across various fields of science and industry. The development of digital technology and increased need for online communication, especially after COVID-19, has driven the evolution of Digital Twin into highly accurate digital replications of physical objects in the real world. Essentially, Digital Twin is a virtual representation of physical objects that can mimic the characteristics, behavior, and functions of the original objects. In the music industry context, Digital Twin enables the creation of realistic virtual representations of musicians and real-time audience interaction in a digital concert performance that can be accessed without time and place limitations. Jin et al. (2022) observed that virtual worlds have become ideal venues for hosting concerts and music festivals, with continuously evolving Digital Twin technology becoming a key component in metaverse concert development [6]. Implementation of Digital Twin in the music industry is supported by integrating various technologies such as Augmented Reality (AR) and Virtual Reality (VR), enabling users to interact naturally in the virtual world [6]. Jin et al. (2022) noted that although metaverse concerts currently require lengthy production processes, the development of photorealistic avatar technology like Metahuman has opened new opportunities in the digital music industry. Metahuman avatars can become highly accurate digital representations of musicians, allowing them to perform virtually with natural facial expressions and body movements. The presence of photorealistic avatars not only enhances the visual quality of virtual performances but also enables deeper interaction between musicians and audiences in the metaverse environment, opening a new paradigm in how music is produced, distributed, and enjoyed. Implementation of Digital Twin in the music industry is not only focused on virtual performances but also opens new opportunities in digital content monetization. Saputera et al. (2024) explained that blockchain technology integration, especially through NFTs, allows musicians and music producers to create unique and valuable virtual experiences. NFTs can function as ”digital tickets” that provide access to physical concerts as well as exclusive virtual experiences, creating a new business model in the music industry.

C. *Avatar Creation Technology*

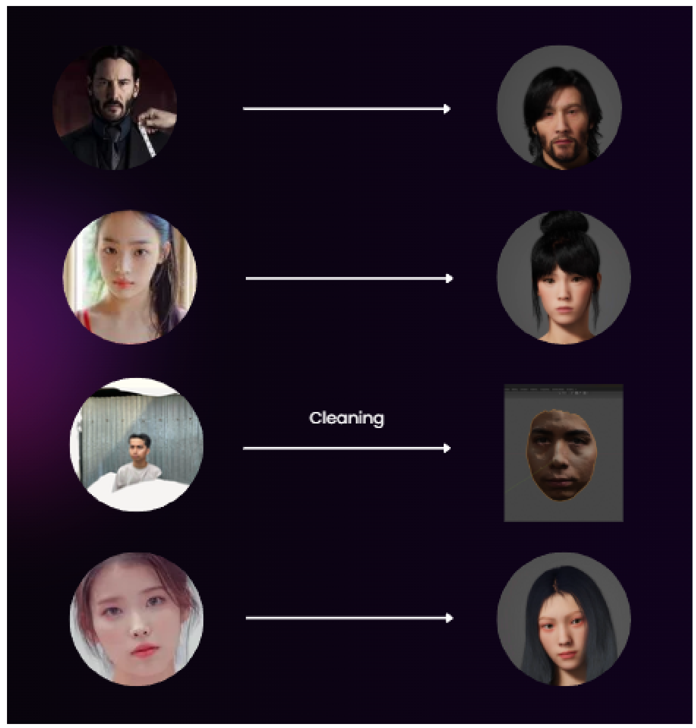

Fig. 1. Digital Twin Metahuman

1) Supporting Tools for Avatar Creation: In developing digital characters using Metahuman Creator, the initial stage begins with creating a facial mesh that can be done using tools like Polycam or HyperHuman AI. After the 3D facial mesh is successfully created, the next process is to use the Metahuman Identity feature integrated in Unreal Engine. This tool can analyze the imported facial mesh and identify key facial features to be transformed into a compatible Metahuman structure. The identification process involves matching facial landmark points and mapping the mesh topology into the standard Metahuman format. The result of this process is a base Metahuman that already has the basic characteristics of the scanned original face, complete with rigging and deformation

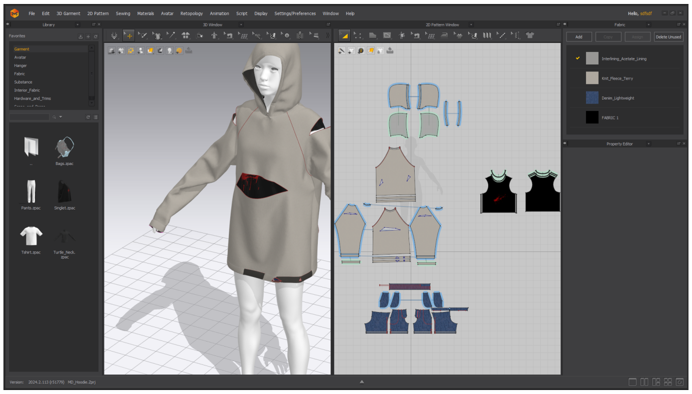

Fig. 2. Make A Cloth Asset Using Marvelous Designer

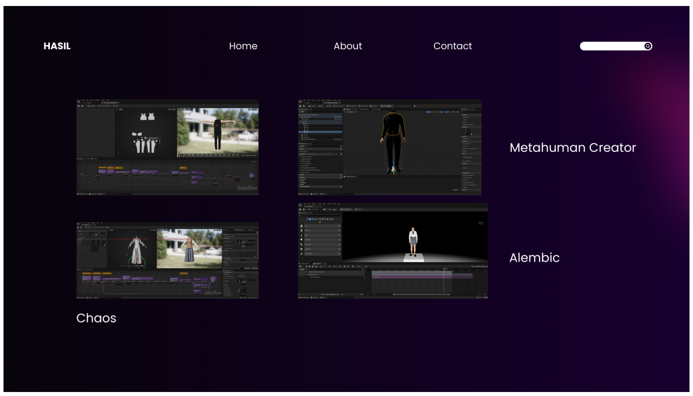

Fig. 3. Cloth Assets In UE5

systems ready for further customization. Metahumans generated from the identification process can then be imported into Metahuman Creator for further refinement and customization. At this stage, users can make detail adjustments such as skin texture, eye shape, facial bone structure, and other elements to achieve a higher level of similarity with the original reference. This process allows developers to create more accurate and personalized digital representations compared to using only the default templates available in Metahuman Creator.

2) Metahuman Creator: Metahuman Creator is a web browser application that was formed by American software developers and video game publishers in February 2021. The application allows developers and creators to create a photorealistic digital character of a human from body, face, hair to clothing equipped with rigging features in a matter of minutes. Metahuman Creator runs in the cloud via Unreal Engine Pixel Streaming which can create high-quality digital characters without requiring high-level specification hardware [10]. Technically, Metahuman Creator offers a high level of customization and flexibility in character creation. Users can select and modify various hairstyles using strand-based hair or hair cards systems for platforms with lower specifications. The application provides 18 different body proportion types and various clothing options. The visual quality produced by Metahuman Creator has a high level of detail ranging from hair, skin, eyes, and teeth as well as the ability of lighting effects that hit the skin very realistically [10]. The materials and textures displayed are able to produce details such as facial wrinkles due to aging with proper shadow deformation when given motion. After the character is created, users can download assets through Quixel Bridge in a format that already supports motion capture and is complete with Level of Detail (LOD) and of course includes mesh, skeleton, facial rig, animation controls and materials. Further development of Metahuman Creator is supported by integration of advanced technologies from various leading vendors. Collaboration with ARKit, Faceware, JALI Inc., Speech Graphics, and DynamixYz enables implementation of more sophisticated animation systems to produce convincing performances. This technological breakthrough also includes the use of Codec Avatars developed together with Digital Domain, which significantly improves capture capability and digital representation. The digital DNA system implemented in this platform allows very detailed control over facial features, covering aspects such as bone structure, skin texture, and expressions that contribute to a high level of realism.

3) Clothing Design and Simulation: Metahuman Creator has provided Metahuman body and clothing models, but the clothing models provided are very limited. One solution as a tool to overcome this limitation is Marvelous Designer which can create various kinds of clothing that match the body of the Metahuman that has been created. Marvelous Designer is a 2D pattern-based clothing simulation software that allows designers to create digital clothing with a high level of realism. Photos of clothing are taken flat to ensure pattern details and textures can be clearly seen, which are then used as references in creating 2D patterns [11]. This software integrates professional pattern making systems with real-time physics simulation, allowing creation of various types of clothing from simple to complex. In its implementation, Marvelous Designer provides various tools such as Pattern Editor for pattern creation and modification, Simulation Engine for cloth physics calculation, and Garment Arrangement tools to adjust how clothing falls and interacts with avatars. In the context of developing clothing for Metahuman, the initial stage begins by importing the static mesh skeleton from Metahuman into Marvelous Designer. This process requires special attention to scale and proportion settings to ensure that the clothing created will fit the avatar. Pattern Editor is used to create basic patterns that follow the size and shape of the Metahuman body, taking into account anchor points on the skeleton to ensure natural movement. The use of measurement systems in Marvelous Designer helps in ensuring size accuracy, while tools such as Internal Line and Segment Sewing allow creation of complex construction details such as folds, hems, and joints. Implementation of materials and textures on clothing is done through Property Editor which allows adjustment of detailed fabric characteristics. Parameters such as thickness, stretch (elasticity), bend (flexibility), friction, and particle distance affect how the fabric will behave when simulated. UV Editor plays an important role in texture mapping, allowing placement of patterns and visual details with precision. These physical property settings must be adjusted to the type of fabric being represented, for example cotton will have different parameters than silk or leather, affecting how the clothing will move and react to gravity and collision [11].

D. Motion Capture Technology

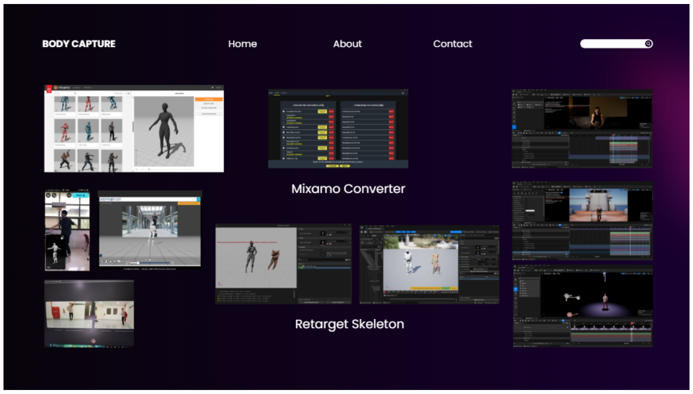

Fig. 4. Body capture from Move One, DeepMotion and Rokoko

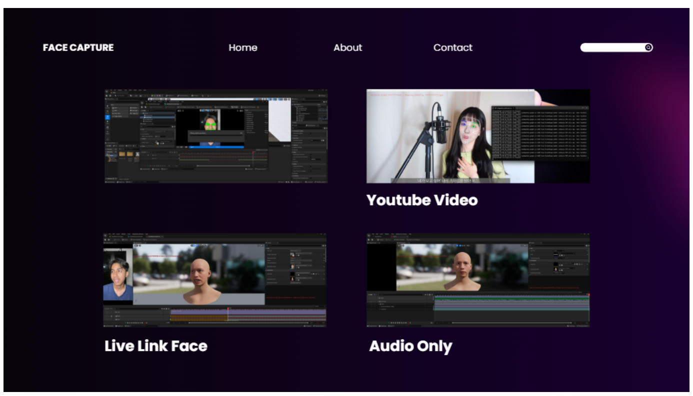

Fig. 5. Facial capture from LiveLink and Face Landmarker Link 02

1) Move AI: Move AI is an artificial intelligence-based motion capture technology developed to simplify the process of capturing human body anatomical movements. This technology allows users to produce high-quality motion capture data using just a camera or smartphone, without requiring special equipment such as motion capture suits or markers. The system uses AI algorithms to analyze videos and extract 3D motion data into skeletons that can be directly applied to 3D characters. Move One is an entry-level service package from Move AI designed for small to medium scale motion capture needs using a single-camera system to create 3D animations based on iPhone. This package is further divided into several types such as Free Plan, Starter Plan and Basic Plan. The author will use Free Plan with a limitation of 30 seconds each month while other package types have a limit of 60 seconds and can produce 3D movement data a maximum of 3 times with exports in the form of FBX from 2D video capture. The single camera capture system on Move One works through several process stages. The first thing users do is record actor video from one viewpoint with attention to several technical requirements to get the best quality. Move One recommends using stable devices, users can use a tripod or lean the device against a wall. After the camera is ready, the actor is required to start with pose-A where the actor stands upright with both arms stretched to the sides with palms facing down, feet aligned with shoulder width, body straight up and head facing forward. To improve accuracy of floor plane estimation, actors are advised to walk around the recording area. Actors must also wear fitting clothes with contrast colors to the background, avoiding loose clothing that can interfere with limb detection. Make sure the video capture area has only one actor visible to the camera as it can affect detection of the desired actor. After users take videos with attention to these guidelines, the file can be uploaded to the Move AI cloud processing where AI algorithms will analyze each frame to detect body poses and extract movement information in 3D space. This algorithm is able to estimate the position and rotation of each joint of the body including hand and finger movements from 2D data captured by a single camera. The processing results can then be downloaded in FBX format which already includes skeleton and animation data that is ready to be applied to Metahuman digital characters in Unreal Engine.

2) DeepMotion: DeepMotion is another AI-driven motion capture solution that complements the motion capture workflow used in this research. Similar to Move AI, DeepMotion enables markerless motion capture through standard video footage but offers additional refinement capabilities specifically beneficial for character animation in game engines. The platform utilizes advanced deep learning algorithms to analyze human movement patterns and translate them into precise skeletal animations. DeepMotion Animator, the specific tool employed in this research, provides powerful post-processing features that address common issues in AI-based motion capture. While Move AI excels at the initial capture and skeleton estimation, DeepMotion Animator offers superior capabilities in motion refinement, including foot sliding correction, movement stabilization, and physics-based constraints that enhance the naturalism of the resulting animations. The workflow integration between Move AI and DeepMotion created a complementary pipeline: initial motion capture data generated through Move AI was imported into DeepMotion for refinement, particularly to correct artifacts in foot placement during dance movements and to enhance the physical believability of weight shifting during performance sequences. DeepMotion's physics-aware algorithms also improved the interaction between the character's body and virtual clothing, resulting in more convincing movement dynamics for the digital twin performances. For complex choreography sequences in the music video showcases, DeepMotion's ability to blend and transition between different movement segments proved especially valuable, allowing for seamless combinations of separately captured dance routines. The refined animations were then exported in FBX format compatible with Unreal

Engine's animation framework, enabling direct application to the Metahuman character rigs with minimal additional adjustment required.

3) Rokoko Dual Vision: Rokoko Vision is the company's AI-based motion capture platform that transforms 2D video into 3D animation data using deep learning and computer vision. Originally branded as "Rokoko Video," it was rebranded as Rokoko Vision in 2024. The system offers two capture modes: a free single-camera option with a 15-second recording limit, and a more accurate dual-camera setup available via a subscription plan. The dual-camera system is a key feature, offering superior depth estimation and reduced occlusion by synchronizing input from two camera angles. This setup enhances reconstruction accuracy by providing multiple perspectives of the subject's motion. Rokoko recommends a minimum recording area of 4×4 meters, producing a usable capture space of approximately 2×2 meters, scalable for larger projects (Rokoko, 2024a). Cameras should be positioned 45°–90° apart, with at least 2 meters of distance from the subject. Full HD (1920×1080p) resolution at 60 fps is advised for optimal results.

Calibration is essential in dual-camera setups and involves the use of printed checkerboard patterns and calibration markers. These must be printed at exact scale on flat, stable surfaces to avoid distortion. Intrinsic calibration ensures spatial accuracy by capturing the checkerboard in at least nine positions per camera (Rokoko, 2024b). Rokoko Vision supports various RGB cameras, including webcams, modern smartphones, DSLRs, and mirrorless cameras, provided they are recognized as input devices by the operating system. DSLR and mirrorless cameras offer superior image quality and are supported when connected via cable or Wi-Fi.

4) Live Link Face: Live Link Face is an application developed by Epic Games to track facial and neck movements in real-time that can be directly integrated with Unreal Engine. Lee (2023) explains that this application allows users to record and transfer facial expressions directly to Metahuman characters through real-time streaming, where the recording results can be saved as files that can be reused for animation process. In terms of performance and accuracy aspects, experimental research shows a comparison between Live Link Face with professional solutions Faceware and Facegood. Live Link Face achieves 100% accuracy for neutral expressions (equivalent to Faceware and Facegood), 75% for happy expressions (compared to Faceware 62% and Facegood 83.34%), and 87.5% for surprised expressions (compared to Faceware 62.5% and Facegood 83.34%) [13]. Although it is a more affordable solution compared to Faceware, Live Link Face is able to produce competitive results, even outperforming professional solutions in some expressions. Live Link Face provides two different capture modes to accommodate various needs and devices. The first mode is Live Link (ARKit) which is the original capture mode using Apple's ARKit framework to produce blendshape data that can be streamed to Unreal Engine. The second mode is Metahuman Animator which requires devices with TrueDepth cameras to record video, depth, and audio data simultaneously. The Metahuman Animator mode produces more comprehensive data because it includes facial depth information, enabling more detailed and realistic animation when processed using the Metahuman plugin in Unreal Engine.

5) Face Landmarker Link 02: Face Landmarker Link 02 is a specialized implementation of Google's MediaPipe Face Landmarker technology designed for real-time detection and tracking of 3D facial landmarks. It identifies up to 468 landmarks across the eyes, nose, mouth, and facial contours, enabling direct integration with blendshape-based 3D animation systems. The system architecture employs two neural networks: a BlazeFace-based face detector optimized for mobile inference, and a 3D face landmark model that predicts surface geometry within detected facial regions. Landmark detection begins with full-frame face localization, followed by refined analysis of cropped regions. Temporal tracking utilizes landmark data from previous frames to enhance stability and reduce computation, while re-detection is triggered when the face is no longer reliably tracked.

Training involves a multi-objective transfer learning strategy, combining synthetic 3D landmark data and annotated real-world 2D contours. This enables generalization across both simulated and real environments. Output consists of a face mesh with 3D vertex coordinates (XYZ), while texture coordinates (UV) and mesh topology are inherited from a canonical face model. Integration with Metahuman allows for natural and responsive facial expression capture, including subtle micro-expressions often missed by conventional systems. Real-time performance enables immediate feedback for iterative adjustments. For optimal accuracy, the system requires at least 720p at 30fps (preferably 1080p at 60fps), even lighting without harsh shadows, and a subject distance of 60–120 cm to minimize distortion and ensure detail preservation. While multi-face tracking is supported, single-face mode provides more stable and precise results for motion capture. Additionally, the system assigns confidence scores to each landmark, allowing for intelligent filtering and improved post-processing.

6) Mediapipe: MediaPipe is an open-source machine learning framework developed by Google Research to build multimodal perception pipelines that process video, audio, and sensor data in real-time. In the domain of computer vision, MediaPipe offers a suite of pre-built solutions, including face mesh detection, which constitutes a core component of this research [14]. MediaPipe adopts a graph-based architecture in which each node represents a calculator executing a specific function. This modular design enables complex data processing with synchronized execution—an essential aspect for motion capture applications requiring high temporal precision. One of its main strengths lies in its cross-platform compatibility and GPU acceleration using OpenGL or Metal.

For facial detection, MediaPipe utilizes BlazeFace [15], a lightweight convolutional neural network optimized for real-time inference on mobile devices. Once a bounding box is detected, a trained neural network is employed for facial landmark estimation. The face mesh model can identify 468 three-dimensional landmarks (x, y, z) that comprehensively map facial anatomy. The x and y coordinates represent positions in the image plane, while the z-axis offers relative depth estimation. All coordinates are normalized to a [0, 1] range to support various input resolutions. The mesh generation involves triangulation and interpolation across landmarks, guided by the topological structure of the human face. Each landmark is accompanied by a confidence score indicating detection reliability. To maintain frame-to-frame consistency, MediaPipe employs a combination of motion prediction and landmark refinement.

In motion capture applications, MediaPipe provides a markerless and camera-only approach, eliminating the need for physical markers or multi-camera setups. This makes it a cost-effective alternative to traditional systems. The 3D facial landmarks produced by MediaPipe are suitable for expression analysis and can be compared to professional blendshape data. Comparative studies show that MediaPipe performs competitively against conventional motion capture systems. Park and Kim [16] report that MediaPipe achieves high correlation in capturing fundamental expressions (e.g., happiness, anger, surprise), although it still faces challenges in recognizing micro-expressions, often requiring additional post-processing.

MediaPipe also demonstrates robustness under adverse conditions such as partial occlusion, extreme lighting, and rapid head movement. Techniques like multi-frame analysis and adaptive thresholding help preserve tracking continuity. From an implementation perspective, MediaPipe offers a comprehensive API and supports export of landmark data in JSON/CSV formats for further analysis and validation against ground-truth data from professional systems. As an actively maintained and evolving framework, MediaPipe provides a solid foundation for facial motion capture research. Its efficiency is further enhanced through model optimization techniques such as knowledge distillation and quantization, enabling real-time applications without compromising geometric fidelity.

E. Unreal Engine 5

Unreal Engine 5 serves as the integration platform that unites various components in creating realistic digital avatars. The integration process begins by importing 3D face models produced from Polycam or HyperHuman AI into Metahuman Creator, where the model will be used as the basis for Metahuman face formation. Through Metahuman Identity and Auto Tracking features, the 3D model will be analyzed to identify important facial features such as eyes, nose, and lips which will then be used to produce Metahuman characters that resemble the original real-world model. This approach enables production of digital avatars in a simple, fast, and efficient manner, which is not only beneficial for the live broadcast content industry, but can also improve quality and diversification of content in various sectors, including independent media, news media, education, and science [14]. In animation aspects, Unreal Engine 5 provides a system that allows combining motion capture data from various sources in one sequence. Body movement data produced by Move AI in FBX format can be imported and applied to Metahuman skeleton through the Control Rig system. Meanwhile, facial movements captured using Live Link Face can be directly integrated in real-time or saved as animation files for use in other things. Both types of animation data can be synchronized in Animation Blueprint to produce unified and natural performances. For clothing aspects, models created using Marvelous Designer can be imported into Unreal Engine 5 as separate meshes which are then attached to the Metahuman character. This engine supports cloth physics simulation which allows clothing to move realistically following character movements. Level of Detail (LOD) and mesh optimization can be set to ensure optimal performance without sacrificing visual quality. All these components can be rendered in real-time by utilizing lighting and material systems available in Unreal Engine 5 to produce photorealistic visual output.

1) Blueprint: Blueprint Visual Scripting in Unreal Engine 5 is a node-based programming system that allows developers to create gameplay elements without writing traditional code. This system uses a visual interface where logic blocks called nodes can be connected to create complex behaviors, particularly useful for implementing interactive features in virtual concert applications. Blueprint allows designers to create dynamic behaviors in real-time by connecting various nodes representing functions, variables, events, and logic controls. The system enables rapid prototyping and iteration of features such as character movement control, interactive environment elements, and response to user inputs within the digital twin concert implementation. With Blueprint's event-driven architecture, developers can trigger specific actions based on user interactions or timeline events during performance sequences. A key advantage of Blueprint is its integration with Unreal Engine's other systems, including Metahuman animations, physics, materials, and rendering. This integration enables seamless control of digital twin performance aspects, such as triggering facial expressions in response to audio cues or coordinating clothing physics with character movements. Blueprint's visual debugging tools allow developers to trace execution flow and monitor variable states during runtime, facilitating efficient troubleshooting of complex interactive features.

III. METHODOLOGY

A. System Design

This research employs a comprehensive methodology involving multiple integrated processes to develop digital twin avatars for virtual music concerts. The approach begins with extensive data collection, gathering visual and motion data of musician subjects through complementary technologies. We utilize HyperHuman AI and Polycam for 3D facial mesh generation, capturing the distinctive features necessary for creating recognizable digital representations. Simultaneously, we collect motion data using Live Link Face for facial expressions and Move AI for body movements, ensuring the avatars can replicate the nuanced performances characteristic of live music concerts. Following data collection, we process the gathered information to create Metahuman avatars with realistic features closely matching the reference subjects. This stage involves importing the 3D meshes into Unreal Engine 5's Metahuman Identity system, which transforms them into fully-featured digital humans with customizable attributes. The system analyzes facial landmarks and structures, creating a digital representation that maintains the unique characteristics of the original subject while leveraging Metahuman's advanced animation capabilities. The design and implementation of realistic clothing represents a critical component of our methodology. Using Marvelous Designer, we create detailed garments customized to each avatar's dimensions and performance requirements. These clothing assets are implemented through two complementary approaches: Alembic geometry cache for pre-calculated deformations and Chaos Cloth physics for real-time dynamic elements. This dual approach balances visual quality with computational efficiency, particularly important for maintaining performance in interactive environments. Facial motion capture integration forms the expressive core of our digital twins. We capture and process facial performances using Live Link Face with the Metahuman Animator mode, which records not only video but also depth data critical for accurate facial tracking. The resulting animations provide the subtle microexpressions and lip synchronization essential for convincing musical performances. Similarly, body motion capture integration using Move One technology captures the characteristic movements and choreography of the performers. This process includes careful retargeting and cleaning to address structural differences between the motion capture data and Metahuman skeletal systems, particularly in complex areas such as hands. The methodology culminates in comprehensive asset integration, combining all components into cohesive digital twin avatars within Unreal Engine 5. This integration phase utilizes Sequencer for timeline management and performance synchronization. Finally, rigorous testing and evaluation assesses visual quality, motion naturalness, and system performance across various scenarios designed to simulate real-world usage conditions in virtual concert environments.

B. Tools and Equipment

The implementation of this research required a carefully selected set of specialized software and hardware tools to achieve the desired level of realism and performance. For software components, we employed HyperHuman AI for generating high-quality 3D facial meshes from single images, complemented by Polycam for more detailed 3D scanning using multiple photographic angles. These tools provided complementary approaches to capturing facial geometry, allowing us to compare and select the most accurate representations. Metahuman Creator served as the central platform for avatar development, providing the sophisticated rigging and animation systems necessary for lifelike digital humans. For clothing design and simulation, Marvelous Designer offered powerful pattern-based garment creation capabilities with physics-based cloth simulation. Body movements were captured using Move One, which leverages computer vision algorithms to extract skeletal motion data from standard video recordings without requiring specialized hardware. Facial expressions and micromovement data were collected through the Live Link Face application, which utilizes the TrueDepth camera system on iOS devices to capture detailed facial performance data. Unreal Engine 5 served as the integration environment, providing advanced rendering capabilities, physics simulation, and the Sequencer tool for timeline management. The hardware configuration centered around an Acer Nitro AN51558 laptop equipped with a 12th Generation Intel® Core™ i9-13900H processor, NVIDIA® GeForce RTX™ 3060 graphics processing unit with 6GB dedicated GDDR6 memory, and 16GB of DDR5 system RAM. This configuration provided sufficient computational power for real-time rendering and physics simulation while maintaining portability for field work. Mobile capture was performed using an iPhone 15 Pro with its advanced 48MP camera system and LiDAR sensor for primary facial and motion capture, supplemented by an iPhone SE 2020 for additional capture angles when required. This hardware selection balanced performance requirements with accessibility, demonstrating that professional-quality digital twins can be created without industrial-grade equipment.

C. Implementation Process

The implementation process for creating digital twin avatars involved multiple interconnected stages, beginning with avatar creation through sophisticated 3D modeling techniques. We employed a dual approach to generating facial meshes, utilizing both HyperHuman AI and Polycam to capitalize on their complementary strengths. HyperHuman AI processed high-resolution frontal photographs through its latent structural diffusion model, rapidly generating detailed 3D facial structures with particular strength in capturing distinctive facial features. Concurrently, Polycam's photogrammetry approach processed up to 70 photographs from multiple angles (the maximum allowed in the free version), creating geometrically accurate 3D scans with excellent spatial precision. These complementary approaches allowed us to compare and select the most accurate representation for each subject. Following initial mesh generation, we performed cleaning operations in Blender, meticulously removing extraneous elements such as hair, neck structures, and clothing to isolate the facial geometry required for Metahuman integration. The processed meshes were then imported into Unreal Engine 5's Metahuman Identity system, which employs sophisticated computer vision algorithms to detect and map key facial landmarks. This system transforms the imported meshes into Metahuman-compatible structures by analyzing 52 distinct facial feature points and mapping them to corresponding control points in the Metahuman framework. Through this process, the system generated complete Metahuman avatars that not only retained the distinctive characteristics of the original subjects but also inherited the advanced animation capabilities of the Metahuman system, including complex blendshapes for facial expressions and sophisticated skin rendering. Motion capture implementation encompassed both facial and body movements, essential for creating convincing musical performances. For facial capture, we utilized the Live Link Face application on iOS devices equipped with TrueDepth camera systems. This technology simultaneously recorded video, depth, and audio data, providing comprehensive information about facial movements during performances. The resulting data underwent processing in Metahuman Animator, which precisely transferred the captured expressions to the created avatars. This process focused particularly on accurately reproducing eye, eyebrow, and mouth movements, which are crucial for conveying emotion and ensuring proper lip synchronization during singing performances. Body motion capture implementation presented unique challenges that required innovative solutions. We employed the Move One system, which processes standard video recordings through advanced computer vision algorithms to extract skeletal movement data. This approach eliminates the need for specialized hardware such as motion capture suits, making the technology more accessible. The system generated FBX animation files containing skeletal movement data, which then underwent retargeting to match the Metahuman skeleton structure using Unreal Engine's IK Retargeter. A significant technical challenge emerged from the structural differences between Move One's simplified skeleton hierarchy and Metahuman's anatomically complete skeleton, particularly evident in hand areas where Move One lacks metacarpal bone representations. We addressed this limitation through a meticulous cleaning process using Metahuman Control Rig Additive, creating supplementary animation layers specifically for the metacarpal bones to achieve natural hand articulation throughout performances. Clothing implementation required balancing visual fidelity with performance considerations, leading us to develop two complementary approaches. For more static or predictable garment elements, we employed Alembic Geometry Cache methodology. This process began with creating detailed garment patterns in Marvelous Designer, precisely tailored to match the dimensions and movement requirements of each avatar. After simulating cloth physics according to the predetermined avatar movements, we exported the results as Alembic cache files containing pre-calculated deformation data. This approach minimized computational overhead during runtime while maintaining high visual quality for clothing elements with consistent movement patterns. To enhance the realism and precision of both body and facial animation, we integrated two specialized motion capture solutions. For body capture, we utilized Rokoko Vision, an AI-driven markerless system capable of converting 2D video input into 3D skeletal data. By employing the dual-camera setup, we significantly improved depth accuracy and reduced occlusion artifacts, particularly during fast or complex movements. Calibration was conducted using checkerboard patterns to ensure accurate spatial alignment across the synchronized video feeds. For facial animation, we implemented Face Landmarker Link 02, a real-time 3D facial tracking system based on Google's MediaPipe technology. This solution enabled the detection of 468 facial landmarks, which were converted into blendshape data compatible with Metahuman facial rigs. Its temporal tracking and high-resolution support allowed us to capture subtle expressions and micro-movements, thereby contributing to a more expressive and believable digital performance. For clothing components requiring more dynamic interaction with the environment, we implemented Chaos Cloth Physics using Unreal Engine 5's native simulation system. This approach involved importing USDA files from Marvelous Designer and configuring sophisticated physics parameters including collision detection, gravity influence, and material properties such as stiffness, bend resistance, and air damping. By carefully tuning these parameters, we achieved realistic clothing behavior that could dynamically respond to character movements and environmental conditions in real-time. This approach provided greater flexibility and realism for elements like loose fabrics and accessories, though it required more computational resources than the Alembic method. The final implementation phase involved comprehensive system integration within the Unreal Engine 5 environment. Using Sequencer, we meticulously synchronized all animation components including body movements, facial expressions, and cloth simulations into coherent performance timelines. We created three distinct music video sequences featuring different choreography, lighting conditions, and environmental settings to demonstrate the versatility of the digital twin avatars across varied performance contexts. The implementation incorporated interactive elements to enhance user engagement, including character swapping functionality allowing transitions between default and Metahuman avatars, level teleportation for seamless environment transitions, and media textures displaying video content on 3D objects within the virtual space. These features collectively created an immersive concert experience that maintained both visual fidelity and interactive responsiveness.

IV. Result and Discussion

A. Visual Quality Assessment

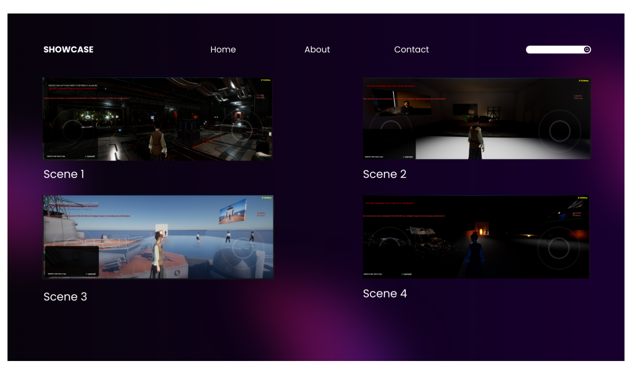

Fig. 6. Assets collection

The resulting digital twin avatars demonstrated high visual fidelity to their reference subjects. The Metahuman system successfully captured distinctive facial features and transferred them to highly detailed 3D models. The integration of skin textures, hair systems, and facial microexpressions created convincing representations capable of conveying emotions during performances.

Fig. 7. Mix Collection

Clothing simulations achieved realistic movement and interaction with character animations. The combination of Alembic cache for static elements and Chaos Cloth for dynamic components provided an optimal balance between visual quality and performance efficiency.

B. Performance Analysis

Performance testing was conducted across five showcase scenes with varying complexity and interactive features. Testing was performed on two different hardware configurations: a mid-range laptop running at "low" scalability settings and a high-performance PC with "cinematic" scalability settings, to evaluate how the digital twin implementation performs across different hardware capabilities.

TABLE I
PERFORMANCE COMPARISON BETWEEN PC AND LAPTOP

Scene	PC Avg	Laptop Avg	Difference	PC Range	Laptop Range
Scene 1	49.63	44.16	1.12x (12.4%)	45-53	40-48
Scene 2	13.07	6.00	2.18x (117.8%)	11-15	5-8
Scene 3	46.65	38.49	1.21x (21.2%)	41-50	3-43
Scene 4	47.38	44.16	1.07x (7.3%)	45-51	40-48
Scene 5	42.12	16.78	2.51x (151.0%)	35-51	10-25

The implementation maintained frame rates above the minimum target of 30 FPS in most testing scenarios on the high-performance PC, with average rates ranging from 42-50 FPS in scenes with Metahuman implementations (Scenes 1, 3, 4, and 5). However, significant performance differences were observed between the two hardware configurations, particularly in scenes with custom assets and high-fidelity Metahuman rendering.

Scenes utilizing pre-optimized assets from Unreal Engine's Featured and Bundled (FAB) Marketplace (Scenes 1, 3, and 4) demonstrated significantly better performance across both hardware configurations, with relatively modest performance differences (7.3%-21.2%) despite the substantial differences in scalability settings. This highlights the critical importance of asset optimization in achieving consistent performance across diverse hardware capabilities.

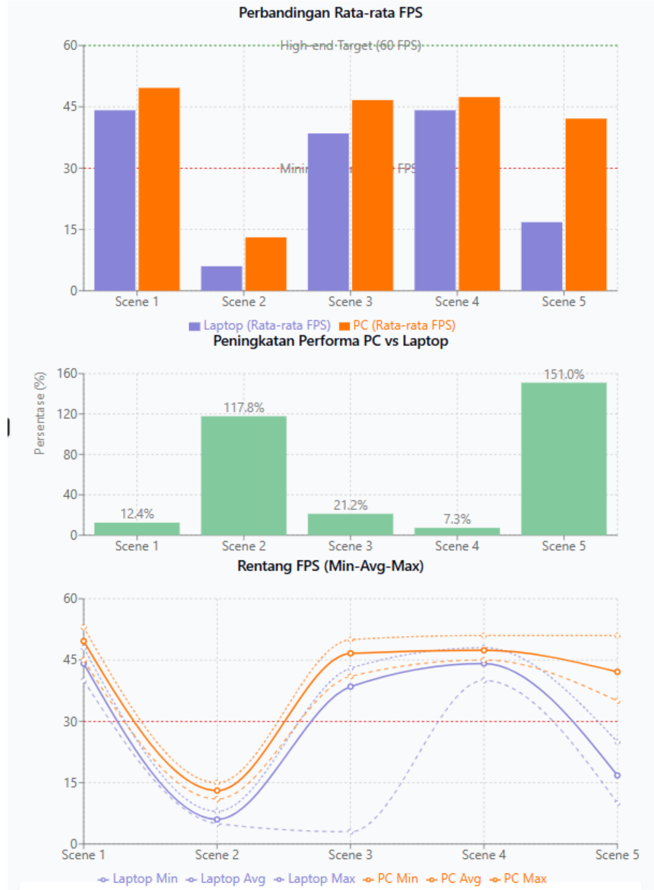

Fig. 8. Performance comparison: (a) Average FPS, (b) Performance improvement, (c) FPS range

In contrast, Scene 2, which implemented custom static meshes and lighting systems, showed poor performance on both configurations, failing to reach the 30 FPS threshold even on high-performance hardware. The 117.8% performance difference between configurations indicates that non-optimized custom assets create significant bottlenecks regardless of hardware capabilities.

Scene 5, which focused specifically on high-detail Metahuman rendering with identical "cinematic" settings on both configurations, revealed the most dramatic performance difference (151.0%). While the PC maintained acceptable performance above 30 FPS (averaging 42.12 FPS), the laptop struggled significantly, averaging only 16.78 FPS. This finding demonstrates that photorealistic character rendering with features such as strand-based hair simulation, subsurface scattering, and high-resolution facial blendshapes is highly hardware-dependent.

Performance optimization strategies including Level of Detail (LOD) management, texture streaming, preloading, and occlusion culling effectively addressed potential bottlenecks in complex scenes with pre-optimized assets. The Unreal Engine scalability system proved effective in maintaining acceptable performance on lower-specification hardware for well-optimized scenes, but showed limitations when dealing with custom-built assets and detailed Metahuman implementations.

C. System Flexibility and Extensibility

The modular approach employed in this research enables the extension of the framework to different performers and musical styles. Once created, the retargeting and animation systems can be applied to new Metahuman avatars without repeating the entire development process, saving significant time and ensuring consistency of movement quality.

The implementation also demonstrates the viability of integrating these digital twin avatars into broader metaverse experiences, particularly those accessible through NFT-based ticketing systems as proposed in the research motivation. However, the performance analysis suggests that scalability considerations must be carefully addressed when targeting platforms with varying hardware capabilities, particularly for implementations featuring high-fidelity Metahuman rendering.

D. Evaluation Of Face Landmarker Link 02

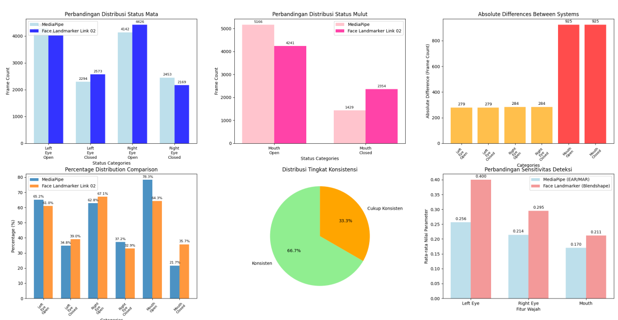

Fig. 9. Mix Collection

A comprehensive evaluation was performed on a dataset comprising 6,595 frames to assess the classification consistency between MediaPipe and Face Landmarker Link 02. Prior to the comparison, an evaluation framework was established to define the agreement methodology and the threshold simplification used in this study. Agreement in this context refers to the degree of classification similarity between the two systems, measured by the absolute percentage difference in binary classification results (OPEN or CLOSED) for each facial movement category. The classification system was derived through the simplification of MediaPipe's original multi-level categories into a binary model, aiming to improve comparison clarity and detection accuracy.

The simplification process involved reducing MediaPipe's multilabel classification—which includes NEUTRAL, CLOSED, SQUINTING, and OPEN for eyes, and CLOSED, SLIGHTLY OPEN, OPEN, and WIDE OPEN for mouth—into a single-threshold binary classification. Specifically, the optimal thresholds were set to 0.1701 for Eye Aspect Ratio (EAR) and 0.1267 for Mouth Aspect Ratio (MAR) in MediaPipe, while Face Landmarker Link 02 used 0.45 for eye blinking and 0.15 for jaw opening. These thresholds were chosen to effectively distinguish between OPEN and CLOSED states while minimizing computational complexity and increasing classification alignment between the two systems. To evaluate consistency, a custom metric was introduced, based on the absolute percentage difference between the status distributions of both systems. The classification outcomes were categorized into four levels of agreement: Highly Consistent for differences of 3% or less, Consistent for differences between 3% and 7%, Moderately Consistent for differences from 7% to 15%, and Poorly Consistent for differences above 15%.

The analysis revealed that eye movement detection showed high levels of agreement between the systems. For the left eye, MediaPipe classified 4,301 frames (65.2%) as open and 2,294 frames (34.8%) as closed, while Face Landmarker Link 02 classified 4,022 frames (61.0%) as open and 2,573 frames (39.0%) as closed. This resulted in an absolute difference of 279 frames, or 4.2%, indicating a consistent level of agreement. A similar pattern was observed in the right eye, where MediaPipe classified 4,142 frames (62.8%) as open and 2,453 (37.2%) as closed, compared to Face Landmarker Link 02's classification of 4,426 frames (67.1%) as open and 2,169 (32.9%) as closed, resulting in a 4.3% difference. These results confirm that both systems are highly reliable in detecting eye movements, with the binary thresholds—EAR 0.1701 and blink 0.45—effectively calibrated for the anatomical characteristics of Asian subjects, which were predominant in this dataset.

In contrast, mouth motion detection exhibited more substantial divergence between the two systems. MediaPipe classified 5,166 frames (78.3%) as open and 1,429 frames (21.7%) as closed, whereas Face Landmarker Link 02 identified 4,241 frames (64.3%) as open and 2,354 (35.7%) as closed. The absolute difference of 925 frames yielded a percentage difference of 14.0%, falling within the moderately consistent category. This discrepancy can be attributed to fundamental methodological differences. MediaPipe relies on a geometric approach using the MAR metric, which is sensitive to subtle variations in lip movement and captures nuanced changes in overall mouth shape. In contrast, Face Landmarker Link 02 uses a blendshape-based JawOpen parameter that emphasizes more substantial jaw movement, focusing on more visually prominent mouth openings. These methodological distinctions represent a trade-off between sensitivity and robustness. MediaPipe, due to its geometric calculations, offers high sensitivity to subtle facial movements, making it well-suited for detailed facial expression analysis. However, this same sensitivity may result in false positives from minor, visually insignificant variations. On the other hand, the machine learning-based approach of Face Landmarker Link 02 yields more stable and visually consistent results but may underperform in detecting micro-expressions or nuanced changes.

The overall distribution of consistency levels showed that four out of six facial movement categories (66.7%) achieved either highly consistent or consistent agreement (below 7%), while the remaining two categories (33.3%) demonstrated moderate consistency with differences ranging from 10% to 15%. The high agreement observed in eye detection suggests that both systems are well-optimized for handling anatomical variability in the eyes of Asian subjects, and that the calibrated thresholds are effective in achieving reliable eye-state classification. The greater variance in mouth detection, however, highlights the need for careful consideration in hybrid system development, where combining geometric and blendshape-based approaches could potentially improve performance in real-time facial animation systems. The findings emphasize the importance of balancing detection sensitivity and robustness when designing facial motion capture pipelines for production environments.

V. Conclusion

This study has demonstrated the feasibility of utilizing Unreal Engine 5's Metahuman technology to create high-fidelity digital twins of musicians, capturing both visual and kinetic realism suitable for virtual concert environments. By integrating facial motion capture (via Live Link Face) and full-body motion capture (using Move AI, DeepMotion, and Rokoko), the resulting avatars were capable of accurately mimicking the facial expressions and bodily movements of simulated performers throughout the duration of a complete musical piece.

Quantitative evaluation of facial blendshape positions obtained from Face Landmarker Link 02, validated through MediaPipe analysis, revealed a high degree of consistency in eye motion detection, with absolute differences between 4.2

The integration of clothing simulation through Marvelous Designer and Chaos Cloth added significant visual depth without compromising performance. Benchmark testing confirmed that the system maintained stable real-time operation above 30 FPS at 1080p resolution, while preserving validated accuracy in facial motion capture, thereby meeting the standards required for delivering immersive user experiences. These findings highlight new opportunities for digital music innovation, including NFT-based virtual concerts, digital merchandise, and the preservation of musical heritage through expressive, high-fidelity digital twins.

Despite these advancements, the system still demands high-performance hardware to maintain graphical fidelity under maximum settings. Consequently, future work is recommended to explore optimization strategies such as Level of Detail (LOD) management, lighting simplification, and fine-tuning of blendshape thresholds to balance visual realism with system efficiency. Overall, this research provides a technically validated solution for the creation of realistic musician avatars and establishes a foundation for metaverse-driven developments in the entertainment industry, ensuring a reliable standard of accuracy and performance.

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